

SP-19 Phase 5: Deep Extensions — $Q(\zeta_7)$, 11/8, Mason-Stothers, Donaldson

Scoped by: Keeper (May 13, 2026) Directive: Casey — “I want to do these next. Setup the scope and board.”
Status: SCOPED, starting May 14

Investigation D: $Q(\zeta_7)$ — The BST Cyclotomic Field

Why this is the most surprising direction

$Q(\zeta_7)$ is where imaginary and real quadratic fields MEET: - $[Q(\zeta_7) : Q] = g - 1 = C_2 = 6$ - $\text{Gal}(Q(\zeta_7)/Q) = (Z/gZ)^*$ has order $C_2 = 6$ - Contains $Q(\sqrt{-7})$ = our CM field (imaginary quadratic) - Contains $Q(\cos(2\pi/7))$ = maximal real subfield, degree $N_c = 3$ - Class number $h = 1$ - Discriminant $= g^{C_2-1} = 7^5 = 16807$

If the $N_c = 3$ units of $Q(\cos(2\pi/7))$ are BST-structured (like $\epsilon_7 = \text{rank}^{N_c} + N_c \sqrt{g}$ was), the entire Kronecker-Weber theorem for $g = 7$ lives in BST integers.

Deliverables

Toy D1 (Lyra + Elie): $Q(\zeta_7)$ complete arithmetic. - Degree $= C_2$, disc $= g^{C_2-1}$, class number 1 - All 3 fundamental units of $Q(\cos(2\pi/7))$ — are coefficients BST integers? - Regulator of $Q(\cos(2\pi/7))$ — BST expression? - Gauss sum: $g(\chi_{-7}) = i \sqrt{g}$. Verify for all characters mod g . - Jacobi sums: $J(\chi^a, \chi^b)$ for χ = primitive character mod g - Decomposition of primes p in $Q(\zeta_7)$: residue degrees and BST patterns - Dedekind zeta: $\zeta_{Q(\zeta_7)}(s) = \text{product over } \chi \text{ mod } g \text{ of } L(s, \chi)$ - Product of $L(1, \chi)$ for all non-trivial χ mod g — class number formula - Target: 25-30 tests

Toy D2 (Lyra): Eisenstein series and class field generators. - $E(f, s, P_2)$ residues at special points on D_{IV}^5 - At Wallach point $s = \rho_2 = 3/2$: gives $L(1, \chi_{-7}) = \pi/\sqrt{g}$ (known) - At other special s -values: do residues give units of $Q(\cos(2\pi/7))$? - Hecke L -functions $L(s, \chi)$ for all χ mod g evaluated at $s = 1$ - Question: does the D_{IV}^5 Eisenstein machinery produce class field generators? - Target: 20-25 tests

Toy D3 (Elie): Gross-Stark conjecture for $Q(\sqrt{7})$ at $p = g$. - p -adic Stark units: Dasgupta-Kakde-Ventullo (2018) proved Gross-Stark - For $p = g = 7$: compute the 7-adic L -function $L_7(s, \chi_7) - L'_7(0, \chi_7) = -\log_7(u)$ where u is the Stark unit - Verify: is $\log_7(\epsilon_7) = \log_7(8+3\sqrt{7})$ a BST expression? - 7-adic valuation of the Stark unit: $v_7(\epsilon_7 - 1)$ — BST integer? - Compare: archimedean Stark (B1, Toy 2175) vs p -adic Stark (this toy) - Target: 18-22 tests

Investigation E: 11/8 Conjecture $= p(C_2)/2^{N_c}$

The connection

Furuta's 11/8 conjecture: For a closed spin 4-manifold M , $b_2(M) \geq (11/8) * |\text{signature}(M)|$

Note: $11/8 = c_2(Q^5) / 2^{N_c} = p(C_2) / 2^{N_c}$

This connects: - $c_2(Q^5) = 11$ (second Chern number, Toy 2176) - $p(C_2) = p(6) = 11$ (Arthur's multiplicity, Toy 2164) - $2^{N_c} = 8$ (instanton step size, Toy 2174)

Also: Rokhlin's theorem says signature divisible by $16 = 2^{\text{rank}^2}$.

Deliverables

Toy E1 (Elie): 11/8 from D_{IV}^5 spectral data. - b_2 and signature for compact quotients of D_{IV}^5 - Hirzebruch signature theorem: $\sigma = L\text{-polynomial}(p_1, p_2, \dots)$ - For D_{IV}^5 : $p_1 = N_c$ (Toy 2176). What is p_2 ? -

Check: does the BST spectral gap force $b_2 \geq (11/8)|\sigma|$ for D_{IV}^5 quotients? - Rokhlin: σ divisible by $16 = 2^{\text{rank}^2}$. Verify for D_{IV}^5 quotients. - The ratio $11/8$: decompose as $p(C_2)/2^{N_c}$. What makes this the optimal bound? - Connection: 11 particle types (Arthur) constrain $8 = 2^{N_c}$ instanton steps - Target: 22-25 tests

Toy E2 (Elie + Lyra): Donaldson-Freedman landscape in BST integers. - Freedman: topological classification by intersection form (rank + signature) - Donaldson: definite forms must be diagonal (smooth constraint) - Known: $E8 + E8$ is topologically realizable but NOT smoothly - $b_2(E8) = 8 = 2^{N_c}$, $\sigma(E8) = -8$. Ratio = $1 < 11/8$ = violation! - $b_2(E8 + E8) = 16 = 2^{\text{rank}^2}$, $\sigma = -16$. Ratio = $1 < 11/8$. - K3 surface: $b_2 = 22 = 2(c_2(Q^5))$, $\sigma = -16$. Ratio = $22/16 = 11/8$ EXACTLY. - K3 saturates the $11/8$ bound! And $b_2(K3) = 2c_2(Q^5)$. BST integers. - Survey: which 4-manifolds saturate or approach $11/8$? - Target: 20-25 tests

Investigation F: Mason-Stothers (Polynomial ABC)

Why this matters

Mason-Stothers theorem (1983): For coprime polynomials $a(t) + b(t) = c(t)$, $\max(\deg a, \deg b, \deg c) \leq \deg(\text{rad}(abc)) - 1$

This is the PROVED function-field analog of ABC. It's elementary (one page proof using the Wronskian). If BST can derive it from D_{IV}^5 structure, the function field ABC is internal to BST.

Deliverables

Toy F1 (Elie): Mason-Stothers via BST. - The classical proof uses: Wronskian $W(a,b) = a'b - ab'$ - $W = 0$ iff a, b linearly dependent (over constants) - Key: $\deg W \leq \deg(a) + \deg(b) - 1$ (derivative drops degree by 1) - But $W = a'c - ac'$ (since $a+b=c$), so W has roots at all common roots of a', c' - BST angle: polynomials over $F_q[t]$ where $q = p^k$ - For $q = 7 = g$: does the Mason-Stothers bound tighten? - For $q = 2 = \text{rank}$: the F_2 case (binary, simplest) - Wronskian as a boundary operator — connection to D_{IV}^5 boundary? - Target: 18-22 tests

Toy F2 (Lyra): Szpiro over function fields. - Szpiro's conjecture for elliptic curves $E/F_q(t)$: $\deg(\Delta) \leq 6\deg(N) + \text{const}$ - For $F_g(t) = F_7(t)$: does the bound tighten to $C_2 \deg(N)$? - The 49a1 analog over $F_7(t)$: construct and compute Szpiro ratio - Compare: number field $\sigma = 3/2$ vs function field σ - Function field BSD: rank = order of vanishing (PROVED by Grothendieck et al.) - If function field $\sigma = 3/2$ too: the ratio is geometric, not arithmetic - Target: 18-22 tests

Investigation G: Donaldson Invariants as Modular Forms

The connection

Gottsche's conjecture (proved): The generating function of Donaldson invariants for an algebraic surface S is a modular form. The weight depends on $b_+(S)$.

If b_+ hits BST values, the modular forms are ones we know from the Eisenstein machinery on D_{IV}^5 .

Deliverables

Toy G1 (Lyra): Donaldson generating functions at BST b_+ values. - $b_+ = 1$ (definite forms): Donaldson invariants are integers, generating function is theta series - $b_+ = \text{rank} = 2$: first nontrivial case. Weight of modular form? - $b_+ = N_c = 3$: weight? Connection to Sym^2 from Bloch-Kato? - The SW basic classes for b_+ odd: Witten's formula connects Donaldson to SW - For K3 ($b_+ = 3 = N_c$): Donaldson invariants are trivial (K3 is hyperkahler) - For CP^2 : $b_+ = 1$, Donaldson invariants fully computed - For $CP^2 \# k \cdot CP^2_{\text{bar}}$: blow-up formula. Does $k = \text{BST integer}$ give special behavior? - Target: 20-25 tests

Work Assignment Summary

Toy	Investigation	Owner	Tests	Dependencies
D1	$Q(\zeta_7)$ arithmetic	Lyra + Elie	~28	None
D2	Eisenstein class field	Lyra	~22	D1
D3	Gross-Stark p-adic	Elie	~20	B1 (done)
E1	11/8 from spectral data	Elie	~24	C1, C2 (done)
E2	Donaldson-Freedman landscape	Elie + Lyra	~22	E1
F1	Mason-Stothers via BST	Elie	~20	None
F2	Szpiro over function fields	Lyra	~20	A1 (done)
G1	Donaldson modular forms	Lyra	~22	C2 (done)

Parallel start: D1, D3, E1, F1, F2, G1 can all launch simultaneously. Second wave: D2 (after D1), E2 (after E1).

Total: 8 toys, ~178 tests estimated. Four investigations.

Success Criteria

Investigation	“Breakthrough” if...	“Strong result” if...
$Q(\zeta_7)$	Units of $Q(\cos 2\pi/7)$ are BST-structured AND Eisenstein produces them	Units are BST, Eisenstein connection C-tier
11/8	$K3$ saturation $= 2 \cdot c_2(Q^{5/2}) \cdot \text{rank}^2$ derived from spectral gap	BST integers appear in 11/8 landscape, no derivation
Mason-Stothers	Function field ABC derived from D_{IV}^5 boundary operator	Function field Szpiro $= 3/2$ (geometric, not just arithmetic)
Donaldson modular	Donaldson at $b_+ = N_c$ is a form we already know from Eisenstein	BST integers appear in weights/levels, coincidence possible

Key Observation (Keeper)

The most powerful single result would be: **** $K3$ surface saturates the 11/8 bound at $b_2 = 2 \cdot c_2(Q^5) = 22$.****
 If this is derivable from D_{IV}^5 spectral data, it connects Arthur’s multiplicity (11 particle types) to smooth 4-manifold topology (the 11/8 threshold). That would be a genuine new theorem linking number theory to topology through BST.